

10.1

$$2 \frac{di}{dt} + 10i = 10v$$

$$v(t) = 3e^{j5t}, \quad i(t) = \underline{I}e^{j5t}$$

$$2 \left[(j5) \underline{I} e^{j5t} \right] + 10 \left[\underline{I} e^{j5t} \right] = 30 e^{j5t}$$

$$\underline{I} = \frac{30}{10 + j10} = \frac{30}{10\sqrt{2}} \angle -45^\circ$$

$$\underline{i(t)} = \underline{\underline{\frac{30}{10\sqrt{2}} \cos(5t - 45^\circ)}}$$

10.2

Governing differential equation:

$$3 \frac{di(t)}{dt} + 6 i(t) = v_s(t)$$

Represent inputs & outputs as complex exponentials:

$$v_s(t) = 5 e^{j30^\circ} e^{j4t}$$

$$i(t) = \underline{I} e^{j4t} \Rightarrow \frac{di(t)}{dt} = (j4) \underline{I} e^{j4t}$$

Substitute into governing diff. eqn:

$$3 [(j4) \underline{I} e^{j4t}] + 6 [\underline{I} e^{j4t}] = 5 e^{j30^\circ} e^{j4t}$$

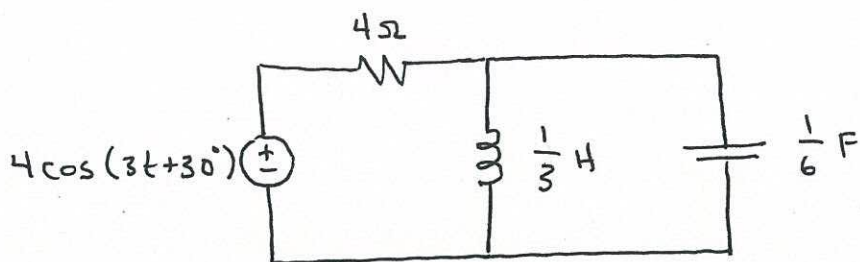
$$[j12 + 6] \underline{I} = 5 e^{j30^\circ}$$

$$\underline{I} = \frac{5 e^{j30^\circ}}{j12 + 6} = \frac{5 \angle 30^\circ}{6\sqrt{5} \angle +63^\circ}$$

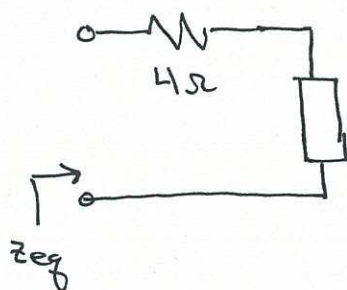
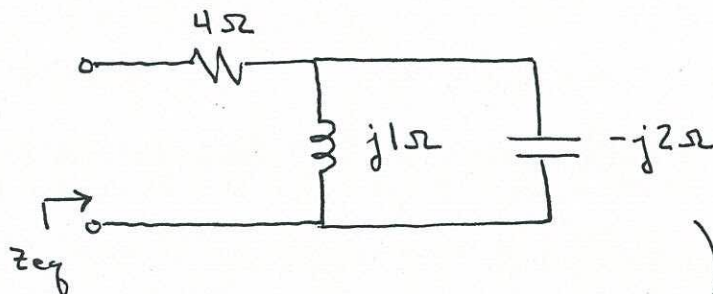
$$\underline{I} = \frac{\sqrt{5}}{6} \angle -33.4^\circ$$

$$i(t) = \frac{\sqrt{5}}{6} \cos(4t - 33.4^\circ)$$

10.3



(a)



$$Z = \frac{(j1)(-j2)}{j1 - j2} = \frac{2}{-j} = j2$$

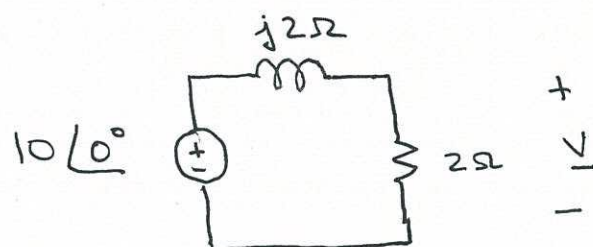
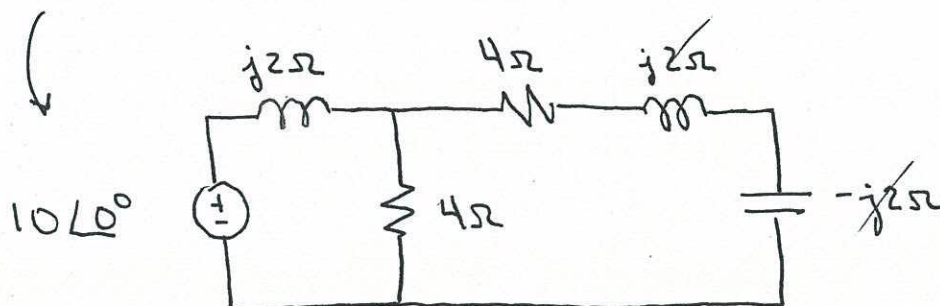
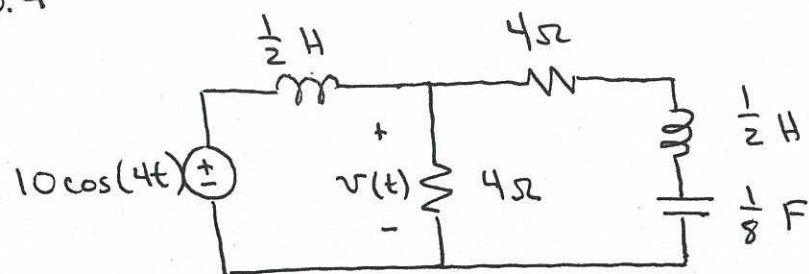
$$\underline{Z_{eq}} = (4 + j2)\Omega$$

$$(b) \quad \underline{I_c} = \underbrace{\underline{V_s} \left(\frac{1}{4 + j2} \right)}_{\underline{I_s}} \underbrace{\left(\frac{j1}{j1 - j2} \right)}_{\text{current divider}} = 4 \angle 30^\circ \left(\frac{1}{4.47 \angle 26.6^\circ} \right) (-1)$$

$$\underline{I_c} = \frac{-4 \angle 30^\circ}{4.47 \angle 26.6^\circ} = -0.894 \angle 3.4^\circ$$

$$\underline{i_c(t)} = \underline{-0.894 \cos(3t + 3.4^\circ)}$$

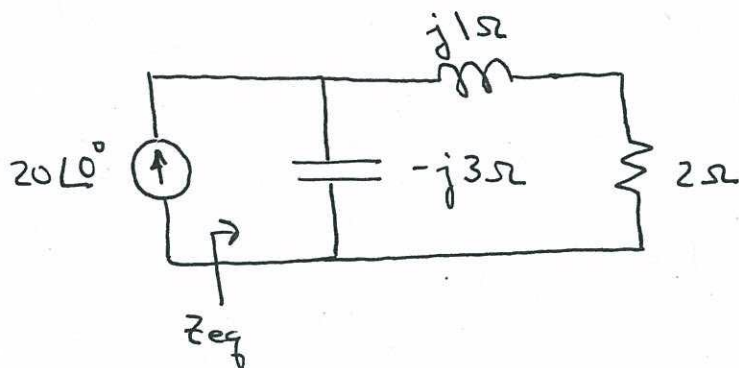
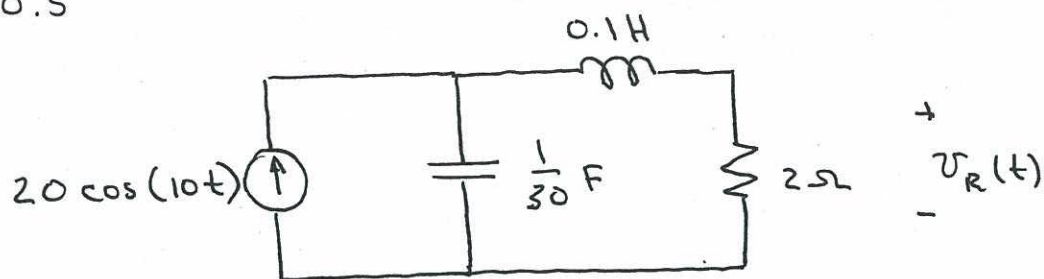
10.4



$$\underline{V} = \frac{2}{2 + j2} (10 \angle 0^\circ) = \frac{20 \angle 0^\circ}{2\sqrt{2} \angle 45^\circ} = \frac{10}{\sqrt{2}} \angle -45^\circ$$

$$\underline{v(t)} = \underline{\underline{\frac{10}{\sqrt{2}} \cos(4t - 45^\circ)}}$$

10.5



(a)

$$Z_{eq} = \frac{-j3\Omega (2+j1)\Omega}{(2+j1)\Omega - j3\Omega} = \frac{(3-j6)\Omega}{(2-j2)} = \frac{3\sqrt{5} \angle -63^\circ}{2\sqrt{2} \angle -45^\circ}$$

$$\underline{Z_{eq} = \frac{3\sqrt{5}}{2\sqrt{2}} \angle -18^\circ}$$

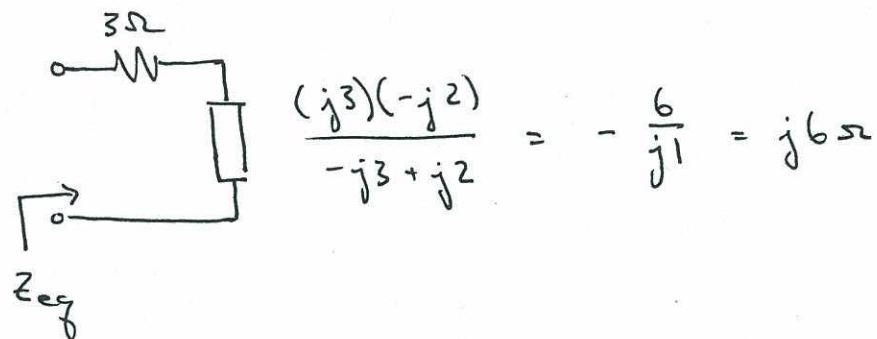
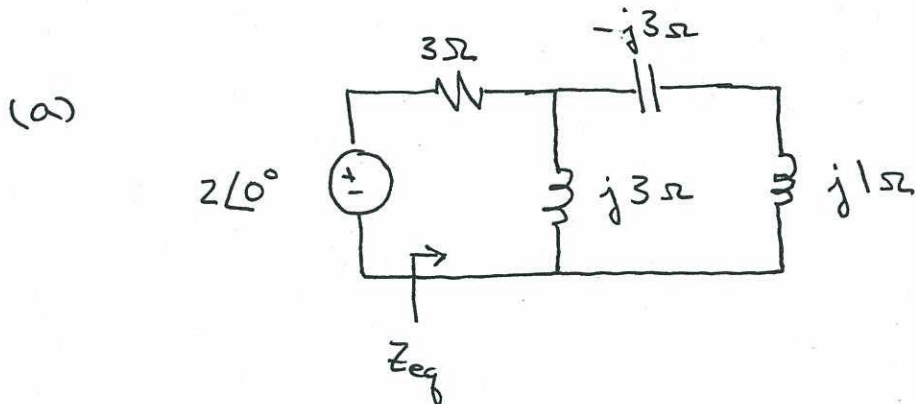
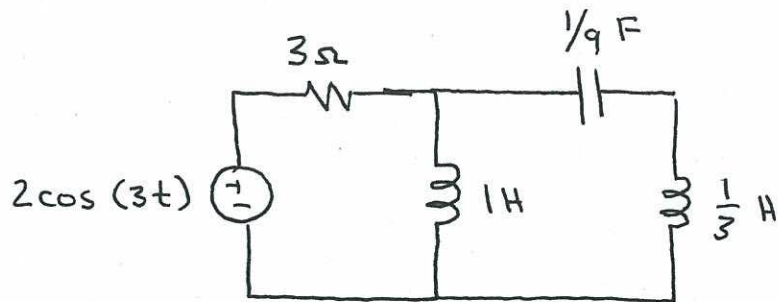
(b) Voltage divider

$$\underline{V_R} = \underbrace{(20 \angle 0^\circ) \left[\frac{-j3 (2+j1)}{2-j2} \right]}_{\text{voltage across current source}} \underbrace{\left(\frac{2}{2+j1} \right)}_{\text{voltage divider}}$$

$$\underline{V_R} = 20 \angle 0^\circ \left[\frac{-j6}{2-j2} \right] = 20 \angle 0^\circ \left[\frac{6 \angle -90^\circ}{2\sqrt{2} \angle -45^\circ} \right] = \frac{60}{\sqrt{2}} \angle -45^\circ$$

$$\underline{v_R(t) = \frac{60}{\sqrt{2}} \cos(10t - 45^\circ)}$$

10.6



$$\underline{Z_{eq}} = (3 + j6) \Omega$$

(b)

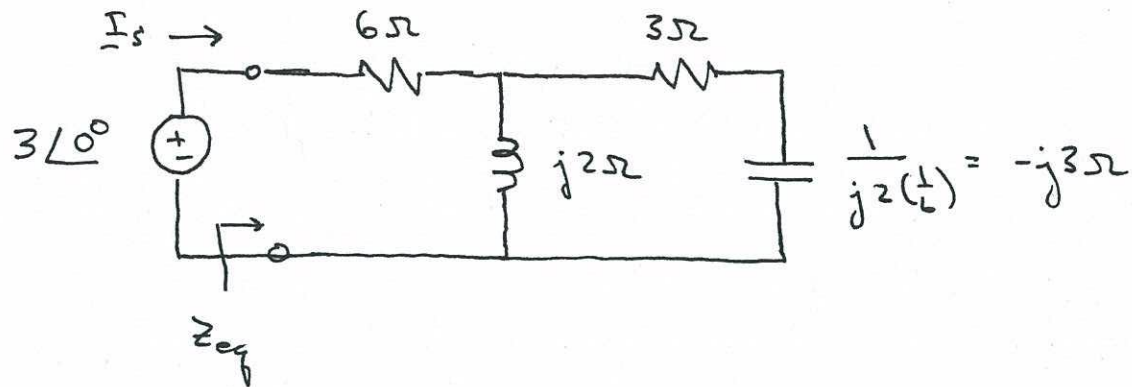
$$\underline{I_s} = \frac{2 \angle 0^\circ}{Z_{eq}} = \frac{2 \angle 0^\circ}{3 + j6} = \frac{2 \angle 0^\circ}{\sqrt{45} \angle 60^\circ}$$

$$\underline{I_s} = \frac{2}{3\sqrt{5}} \angle -60^\circ$$

$$\underline{i_s(t)} = \underline{\underline{\frac{2}{3\sqrt{5}} \cos(3t - 60^\circ)}}$$

10.7

(a) Frequency-domain circuit:



$$Z_{eq} = 6\Omega + \frac{(j2\Omega)(3-j3)\Omega}{j2\Omega + (3-j3)\Omega} = 6\Omega + \frac{j6\Omega + 6\Omega}{(3-j)\Omega}$$

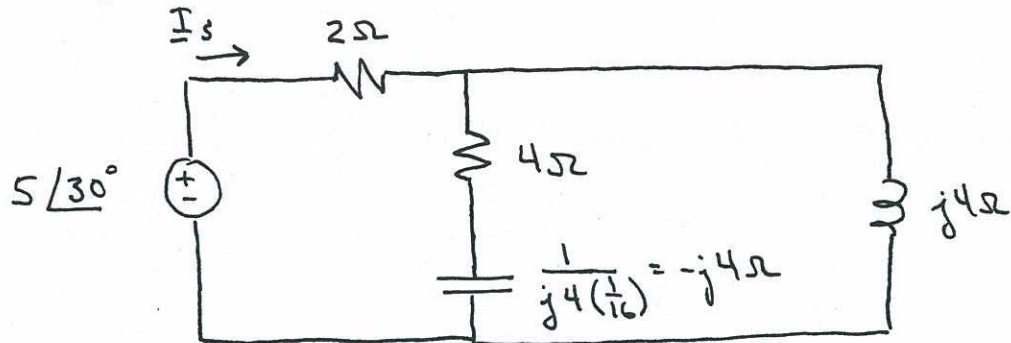
$$Z_{eq} = \frac{18-j6}{3-j} + \frac{6+j6}{3-j} = \boxed{\frac{24}{(3-j)} \Omega = Z_{eq}}$$

$$(b) \quad \underline{I_s} = \frac{\underline{V_s}}{Z_{eq}} = \frac{3\angle 0^\circ}{\frac{24}{3-j}} = \frac{3\sqrt{10}}{24} e^{-j18.4^\circ}$$

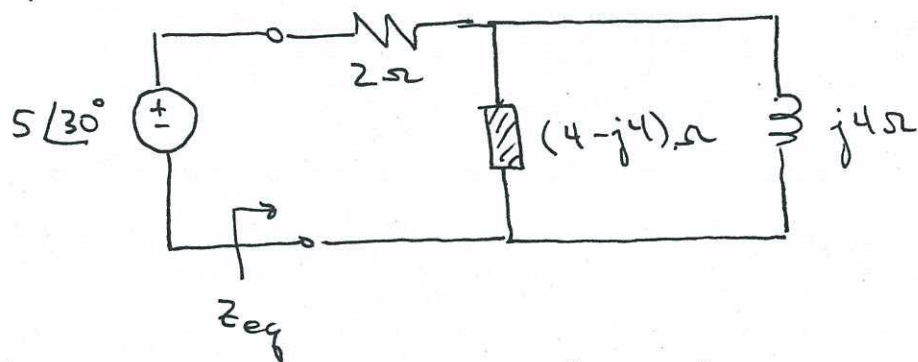
$$i_s(t) = \frac{\sqrt{10}}{8} \cos(2t - 18.4^\circ)$$

10.8

Frequency domain circuit:



(a) Equivalent circuit:



$$Z_{eq} = 2\Omega + \frac{(j4\Omega)(4-j4)\Omega}{j4\Omega + (4-j4)\Omega} = 2\Omega + \frac{j16 + 16}{4}\Omega$$

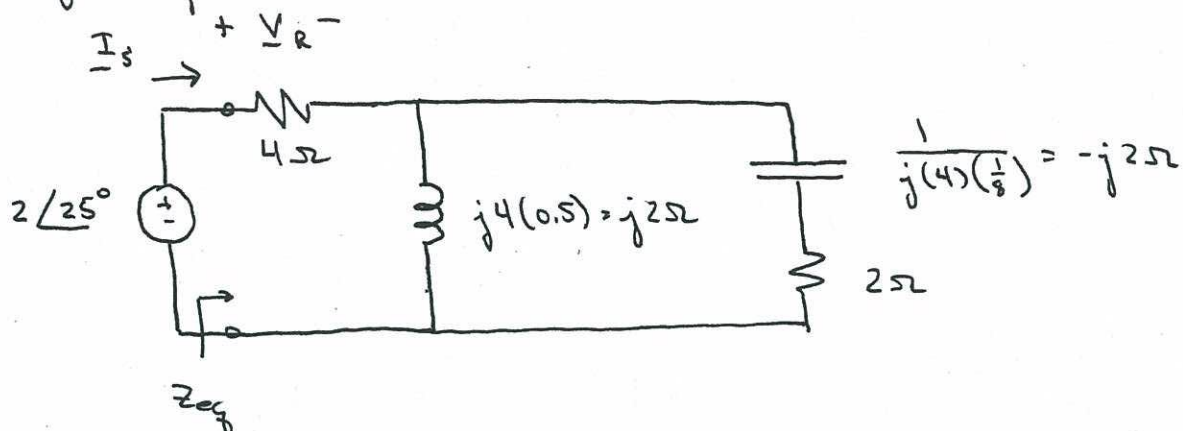
$$Z_{eq} = (j4 + 6)\Omega$$

$$(b) \quad \underline{I}_s = \frac{5\angle 30^\circ}{(6+j4)\Omega} = \frac{5\angle 30^\circ}{2\sqrt{13}\angle 33.7^\circ} = \frac{5}{2\sqrt{13}}\angle -3.7^\circ$$

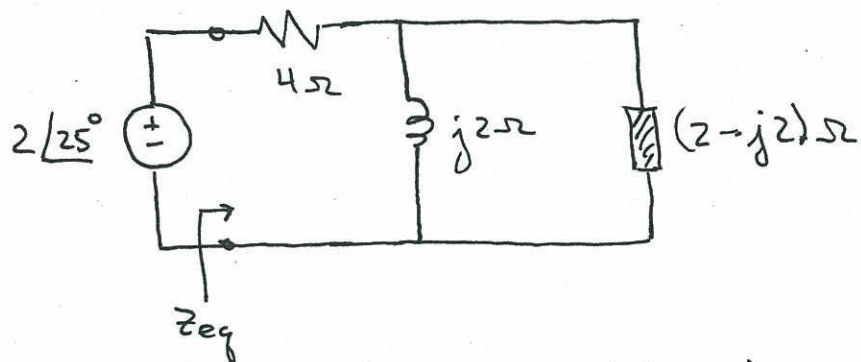
$$i_s(t) = \frac{5}{2\sqrt{13}} \cos(4t - 3.7^\circ)$$

10.9

Frequency domain circuit:



(a) Equivalent circuit:



$$Z_{eq} = 4\Omega + \frac{(j2)(2-j2)}{j2+2-j2} \Omega = (j2+6)\Omega$$

$$Z_{eq} = (j2+6)\Omega = 2\sqrt{10} \angle 18.4^\circ \Omega$$

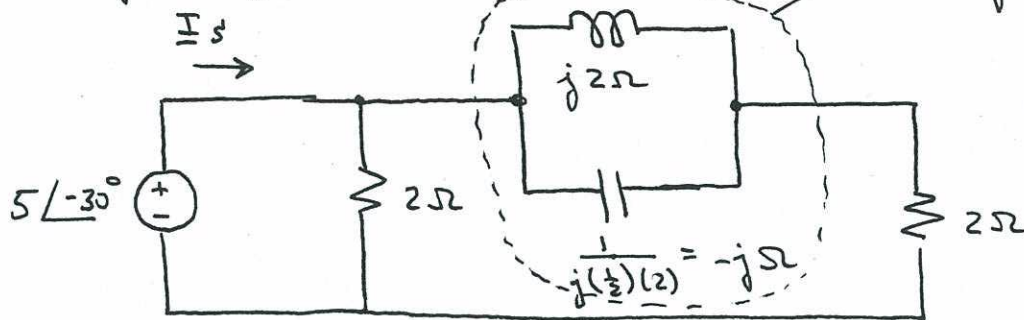
$$(b) \quad \underline{V}_R = (4\Omega)(\underline{I}_s) = (4\Omega) \left(\frac{2\angle 25^\circ}{2\sqrt{10} \angle 18.4^\circ} \right) = \frac{8\angle 25^\circ}{2\sqrt{10} \angle 18.4^\circ}$$

$$\underline{V}_R = \frac{4}{\sqrt{10}} \angle 6.57^\circ$$

$$v_R(t) = \frac{4}{\sqrt{10}} \cos(4t + 6.57^\circ)$$

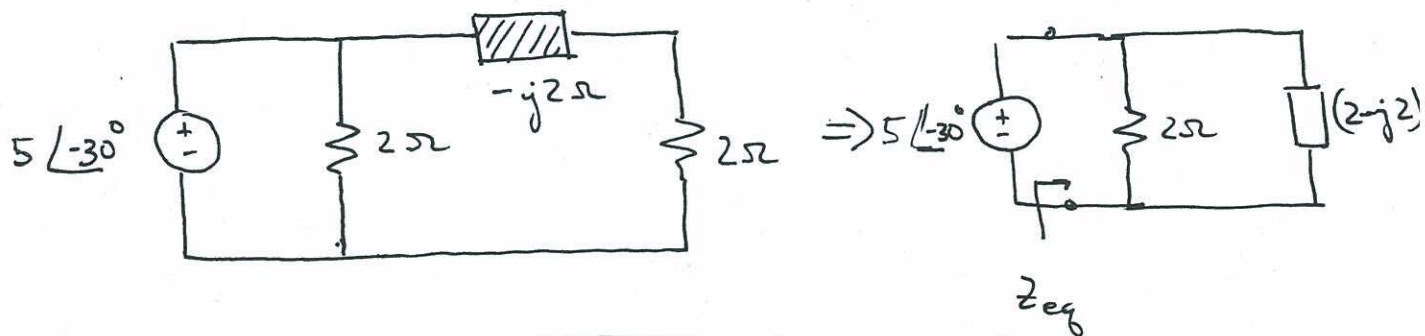
10.10

Frequency-domain circuit:



$$Z_{eq} = \frac{(-j)(j2)}{j2 - j} = \frac{2}{j} = -j2$$

Equivalent circuit:



$$Z_{eq} = \frac{(2\Omega)(2-j2)\Omega}{2\Omega + (2-j2)\Omega} = \frac{(4-j4)\Omega}{(4-j2)}$$

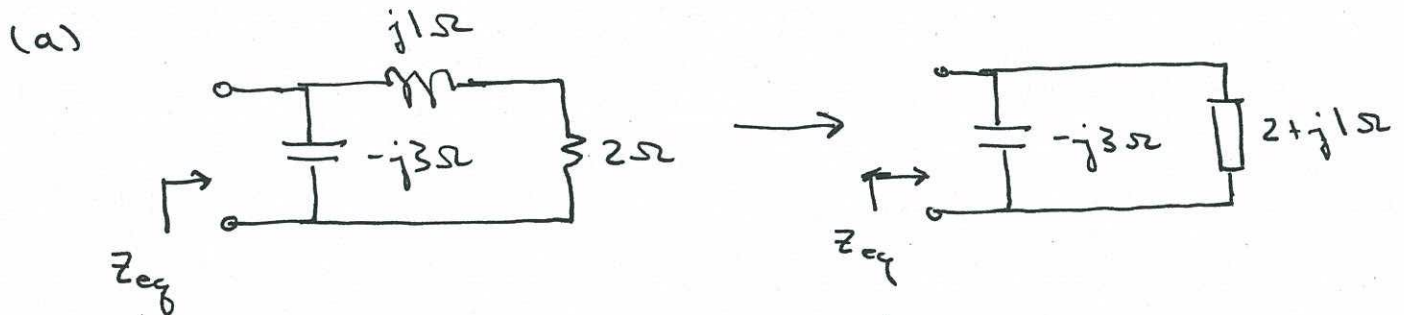
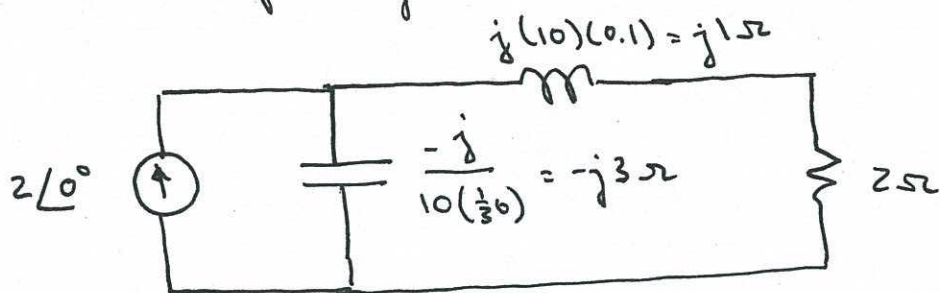
$$Z_{eq} = \frac{2\sqrt{2} \angle -45^\circ}{\sqrt{5} \angle -26.6^\circ} = \frac{2\sqrt{2}}{\sqrt{5}} \angle -18.4^\circ$$

$$\underline{I}_s = \frac{5 \angle -30^\circ}{Z_{eq}} = \frac{5 \angle -30^\circ}{\frac{2\sqrt{2}}{\sqrt{5}} \angle -18.4^\circ} = \frac{5\sqrt{5}}{2\sqrt{2}} \angle -11.6^\circ$$

$$i_s(t) = \frac{5\sqrt{5}}{2\sqrt{2}} \cos(2t - 11.6^\circ)$$

10.11

Frequency domain circuit:



$$Z_{eq} = \frac{(-j3\Omega)(2+j1\Omega)}{-j3\Omega + 2+j1\Omega} = \frac{3-j6}{2+j2} \Omega$$

(b) Probably easiest to use current divider:

$$\underline{I}_R = 2\angle 0^\circ \left[\frac{-j3\Omega}{-j3\Omega + 2\Omega + j1\Omega} \right] = 2\angle 0^\circ \left(\frac{-j3}{2+j2} \right)$$

$$\underline{I}_R = \frac{(2\angle 0^\circ)(3\angle -90^\circ)}{2\sqrt{2}\angle 45^\circ} = \frac{3}{\sqrt{2}} \angle -135^\circ$$

$$\underline{V}_R = (2\Omega) \underline{I}_R \Rightarrow \underline{V}_R = \frac{6}{\sqrt{2}} \angle -135^\circ$$

$$\underline{v}_R(t) = \frac{6}{\sqrt{2}} \cos(10t - 135^\circ)$$